

Introduction to CFD (4RC30)

Chemical reactions

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Outline

Summary of previous lectures

Content of this lecture (Chapter 12)

- Simple chemical reacting system (SCRS)
- Eddy break-up model of combustion

Wrap up

Summary of previous lectures

- **Rate of change + convection = diffusion + (source – sink)**

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- **General form of discretized equations:**

$$a_P \phi_P = \sum a_{nb} \phi_{nb} + S_u$$

$$a_P = \sum a_{nb} + \Delta F - S_P$$

- **Steady flows**

Spatial discretization scheme: CDS, UDS, Hybrid

Algorithm: SIMPLE

- **Unsteady flows**

Time discretization scheme: Explicit, Crank-Nicolson, Implicit

Algorithm: transient SIMPLE, pseudo transient simulations

- **Turbulence models**

k - ε model

Large eddy simulations (LES)

- **Algebraic slip model**

Timescales in reacting flows

- Three timescales (Westerterp, van Swaaij and Beenackers):

$$t_{macro} = \frac{V_r}{\Phi_V}; \quad t_{micro} = \frac{l^2}{12\nu}; \quad t_r = \frac{1}{kc_{A0}^{n-1}}$$

- **Slowest timescale determines reaction rate:**
 - t_{macro} : Simple chemical reacting system
 - t_{micro} : Eddy break-up model
 - $t_{micro} \sim t_r$: Eddy break-up model with reaction
 - t_r : kinetic model (+simple flow model)

Chemical reactions

- **Suppose combustion reaction:**

1 kg of fuel + s kg of oxygen $\rightarrow$ $(1 + s)$ kg of products

- **In mass fractions (Y):**

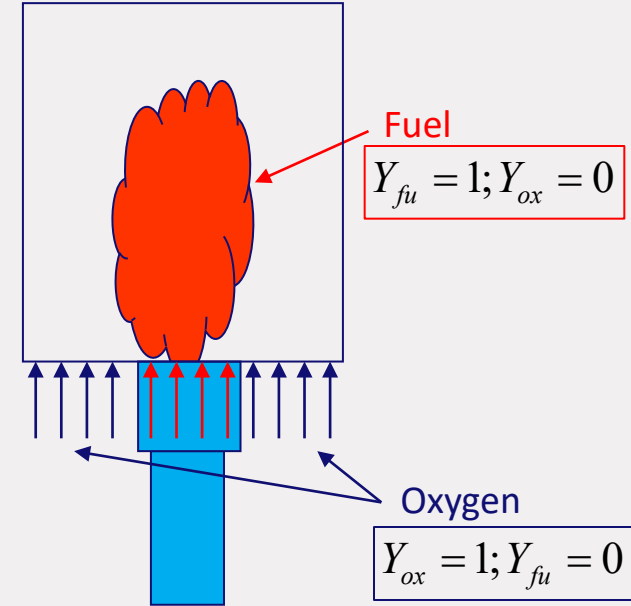
$$Y_{fu} + s \cdot Y_{ox} \rightarrow (1 + s)Y_{pr}$$

- **Total mass fraction:**

$$Y_{total} = Y_{fu} + Y_{ox} + Y_{pr} \equiv 1$$

$$Y = \frac{m_j}{m_{total}}$$

s is stoichiometric $\frac{oxygen}{fuel}$ ratio



Species equations

- Transport equations for mass fractions oxygen and fuel

$$\frac{\partial(\rho Y_{fu})}{\partial t} + \text{div}(\rho Y_{fu} \mathbf{u}) = \text{div}(\Gamma_{fu} \text{grad } Y_{fu}) + S_{fu}$$

$$\frac{\partial(\rho Y_{ox})}{\partial t} + \text{div}(\rho Y_{ox} \mathbf{u}) = \text{div}(\Gamma_{ox} \text{grad } Y_{ox}) + S_{ox}$$

- In order to reduce number of eqs. we assume:

$$\phi = s Y_{fu} - Y_{ox}$$

$$\Gamma_{fu} = \Gamma_{ox} = \Gamma_{\phi}$$

Mixture fraction

- Take *s·fuel* equation and subtract *oxygen* equation:

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma_\phi \text{grad } \phi) + \underbrace{(sS_{fu} - S_{ox})}_{=0}$$

- For supposed stoichiometry:

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma_\phi \text{grad } \phi)$$

- No source term, so ϕ is a so-called passive scalar

Mixture fraction

- Mixture fraction, ξ :

$$\xi = \frac{\phi - \phi_0}{\phi_1 - \phi_0} = \frac{(sY_{fu} - Y_{ox}) - (sY_{fu} - Y_{ox})_0}{(sY_{fu} - Y_{ox})_1 - (sY_{fu} - Y_{ox})_0}; \quad 0 \leq \xi \leq 1$$

Note: in the code
 ξ is called f

- Pure fuel: $Y_{fu,1} = 1; \quad Y_{ox,1} = 0$
- Pure oxygen: $Y_{fu,0} = 0; \quad Y_{ox,0} = 1$
- Substitute to get:

$$\xi = \frac{sY_{fu} - Y_{ox} + Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

Mixture fraction

- **Mixture fraction:**

$$\xi = \frac{sY_{fu} - Y_{ox} + Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

- **Stoichiometric mixture fraction:** $(sY_{fu} - Y_{ox}) \equiv 0$

$$\xi_{st} = \frac{Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

- **Transport equation of mixture fraction:**

$$\frac{\partial(\rho\xi)}{\partial t} + \text{div}(\rho\xi\mathbf{u}) = \text{div}(\Gamma_\xi \text{grad } \xi)$$

What are the mass fractions?

- Suppose you have a given ξ field
- Assume “mixed is burnt” concept
- Derive expressions for the mass fractions:
 - Y_{fu} in terms of: ξ , ξ_{st} and $Y_{fu,1}$
 - Y_{ox} in terms of: ξ , ξ_{st} and $Y_{ox,0}$
 - Y_{pr} in terms of: Y_{fu} and Y_{ox}

$$\xi = \frac{sY_{fu} - Y_{ox} + Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

$$\xi_{st} = \frac{Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

What are the mass fractions?

- The mass fractions for fuel and oxygen, using “mixed is burnt” concept:

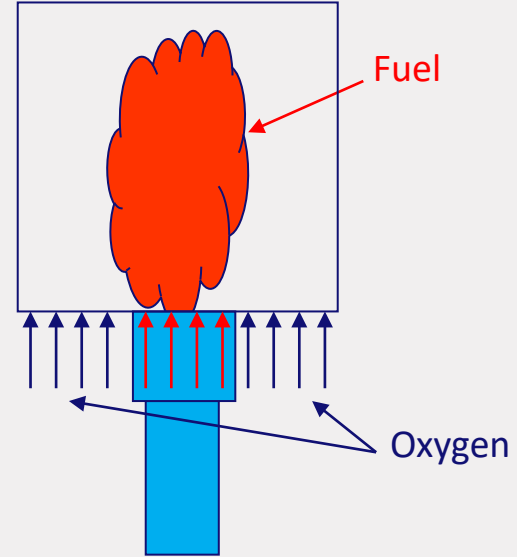
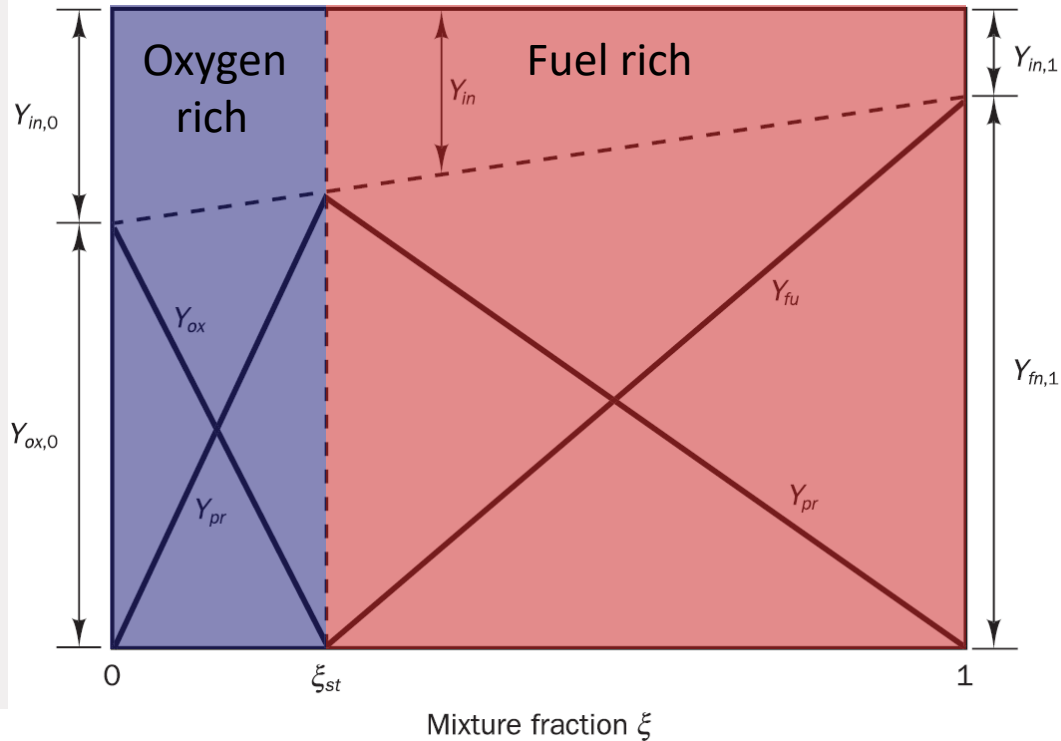
$$\xi_{st} \leq \xi < 1 \quad \Rightarrow \quad Y_{ox} = 0; \quad Y_{fu} = \frac{\xi - \xi_{st}}{1 - \xi_{st}} Y_{fu,1} \quad \text{Fuel rich}$$

$$0 < \xi < \xi_{st} \quad \Rightarrow \quad Y_{ox} = \frac{\xi_{st} - \xi}{\xi_{st}} Y_{ox,0}; \quad Y_{fu} = 0 \quad \text{Oxygen rich}$$

- The mass fractions for the product:

$$Y_{pr} = 1 - (Y_{fu} + Y_{ox})$$

Mass fraction vs. mixture fraction



Simple chemical reacting system (SCRS)

SCRS model:

- PDE for mixture fraction, ξ :

$$\frac{\partial(\rho\xi)}{\partial t} + \text{div}(\rho\xi\mathbf{u}) = \text{div}(\Gamma_\xi \text{grad } \xi)$$

- Algebraic equations for Y_{ox} and Y_{pr}

$$\xi = \frac{sY_{fu} - Y_{ox} + Y_{ox,0}}{sY_{fu,1} + Y_{ox,0}}$$

$$Y_{pr} = 1 - (Y_{fu} + Y_{ox})$$

Temperature

Mixture fraction equation has the same form as the **enthalpy equation** *)

$$\frac{\partial(\rho\xi)}{\partial t} + \text{div}(\rho\xi\mathbf{u}) = \text{div}(\Gamma_\xi \text{grad } \xi)$$

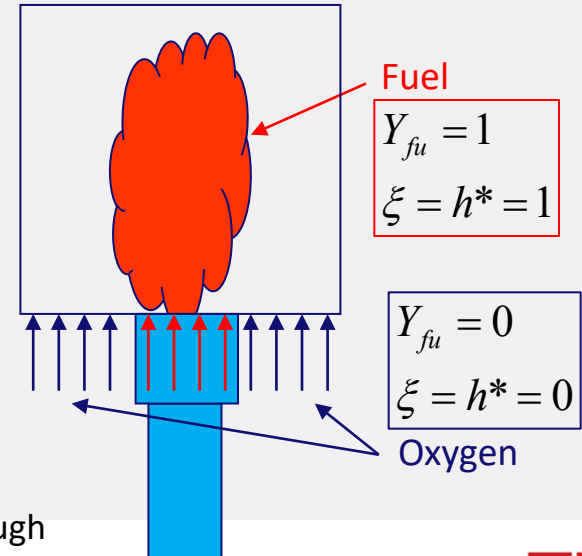
$$\frac{\partial(\rho h)}{\partial t} + \text{div}(\rho h\mathbf{u}) = \text{div}(\Gamma_h \text{grad } h)$$

So, the enthalpy can be related to the mixture fraction and to the temperature for free!

$$\xi = h^* = \frac{h - h_{\text{ox},\text{in}}}{h_{\text{fu},\text{in}} - h_{\text{ox},\text{in}}}$$

$$h = C_p T + Y_{\text{fu}} (\Delta h_{\text{fu}})$$

$$T = \frac{h - Y_{\text{fu}} (\Delta h_{\text{fu}})}{C_p}$$



*) Assuming no heat transport through radiation and adiabatic walls, see p. 373

Micro mixing timescale

- **Micro mixing timescale:**
$$t_{micro} = \frac{l^2}{12\nu} = \frac{\rho l^2}{12\mu_{eff}}$$

- **Eddy length scale:**
$$l = \frac{k^{3/2}}{\varepsilon}$$

- **Effective dynamic viscosity:**
$$\mu_{eff} = \rho C_{\mu} \frac{k^2}{\varepsilon}$$

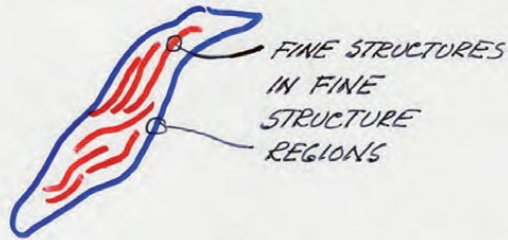
- **Substitution gives:**

$$t_{micro} = t_{eddy} = \frac{1}{12C_{\mu}} \frac{k}{\varepsilon} = \frac{k}{\varepsilon}$$

Eddy Dissipation Concept

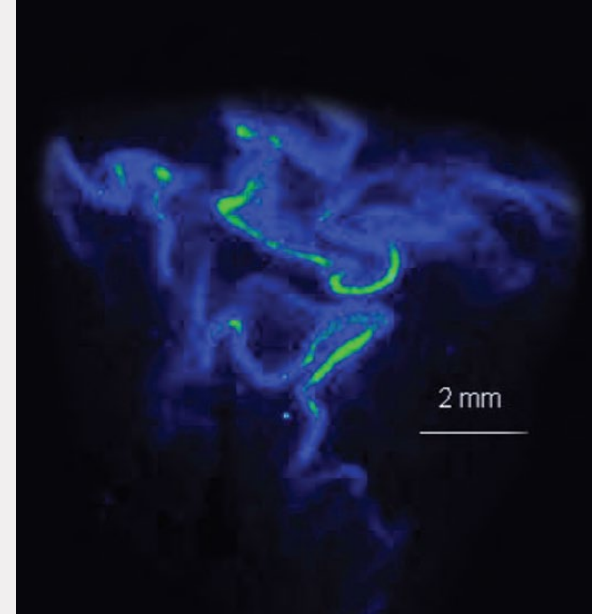
MODELING CONCEPTS

THE EDDY DISSIPATION CONCEPT
OF BFM



STRUCTURES CREATED BY
THE TURBULENCE
DISSIPATIVE STRUCTURES

IGNITION
EXTINCTION
DETAILED CHEMISTRY



Magnussen's original hand sketch of the reactive zones (left), subsequent confirmation using laser photography (right)

Eddy break-up model

- Reaction rate depends on speed of mixing of the species:

$$R_{fu} = -C_R \rho Y_{fu} \frac{\varepsilon}{k}; \quad R_{ox} = -C_R \rho \frac{Y_{ox}}{s} \frac{\varepsilon}{k}; \quad R_{pr} = -C'_R \rho \frac{Y_{pr}}{(1+s)} \frac{\varepsilon}{k}$$

- **Turbulent combustion:** $C_R = 1$; $C'_R = 0.5$
- **Species with lowest speed of mixing determines sink in PDE fuel fraction, Y_{fu} :**

$$S_{fu} = -\rho \frac{\varepsilon}{k} \min \left[C_R Y_{fu}, C_R \frac{Y_{ox}}{s}, C'_R \frac{Y_{pr}}{1+s} \right]$$

Eddy break-up model

Practical tip: it requires non-zero initial values of m_{fu} , m_{ox} and m_{pr} to “ignite” the reaction

$$S_{fu} = -\rho \frac{\varepsilon}{k} \min \left[C_R m_{fu}, C_R \frac{m_{ox}}{s}, C'_R \frac{m_{pr}}{1+s} \right]$$

- Complete model:

- PDEs for mixture fraction ξ and fuel mass fraction Y_{fu} :

$$\frac{\partial(\rho Y_{fu})}{\partial t} + \text{div}(\rho Y_{fu} \mathbf{u}) = \text{div}(\Gamma_{fu} \text{grad } Y_{fu}) + S_{fu}$$

$$\frac{\partial(\rho \xi)}{\partial t} + \text{div}(\rho \xi \mathbf{u}) = \text{div}(\Gamma_{\xi} \text{grad } \xi)$$

- Algebraic equations for other mass fractions, Y_{ox} and Y_{pr} :

$$\xi = \frac{s Y_{fu} - Y_{ox} + Y_{ox,0}}{s Y_{fu,1} + Y_{ox,0}}$$

$$Y_{pr} = 1 - (Y_{fu} + Y_{ox})$$

Eddy break-up model with reaction

- Micro mixing/reaction determine reaction rate
- Finite reaction rate (Arrhenius equation):

$$R_{fu,kinetic} = -A_1 \rho^{-a} Y_{fu}^b Y_{ox}^c \exp(-E / RT)$$

- Minimum of speed of mixing and reaction determines sink in PDE fuel fraction, m_{fu} :

$$S_{fu} = -\min \left[\rho \frac{\varepsilon}{k} C_R Y_{fu}, \rho \frac{\varepsilon}{k} C_R \frac{Y_{ox}}{s}, \rho \frac{\varepsilon}{k} C'_R \frac{Y_{pr}}{1+s}, -R_{fu,kinetic} \right]$$

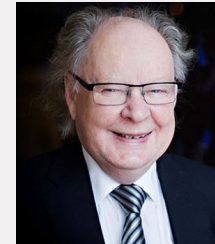
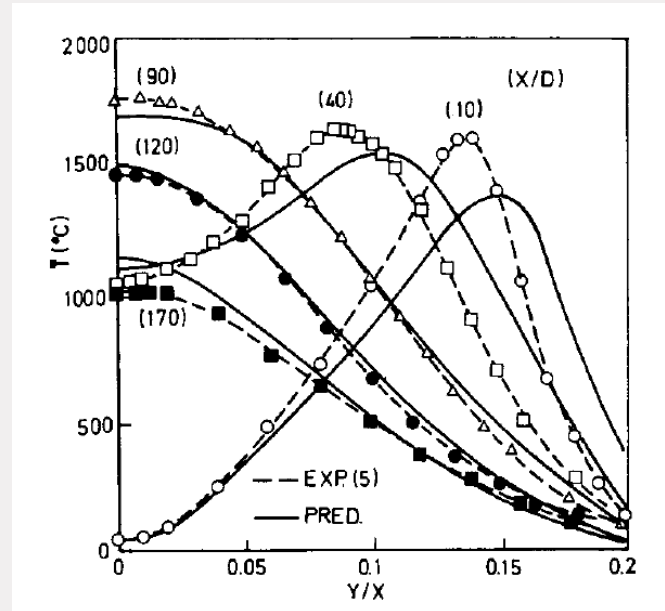
Eddy break-up model with reaction

Complete model:

- Same as standard eddy break-up model
- Note: sink term S_{fu} is different

Example: chemically reacting flow

Results of the eddy break-up model: temperature field in furnace

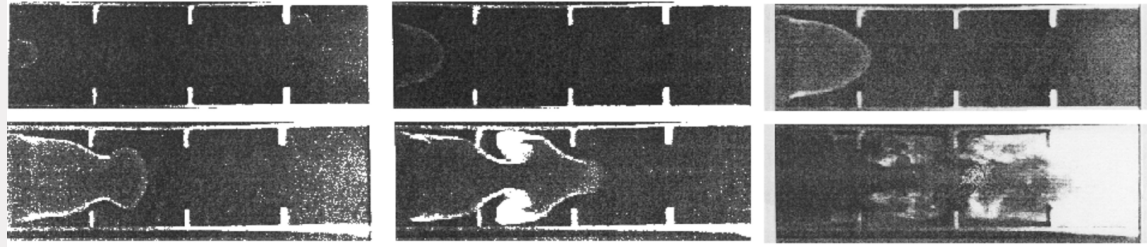


Magnussen and Hjertager, 1976

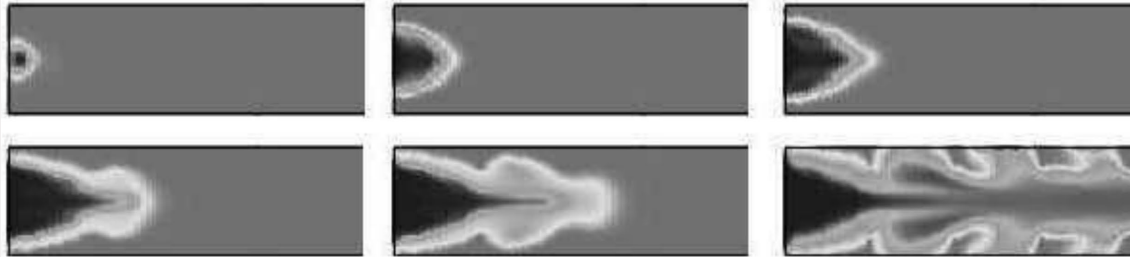
Example: chemically reacting flow

Explosion of premixed propane/air

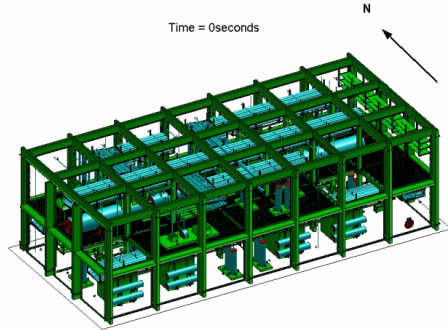
- Experiments (product fraction)



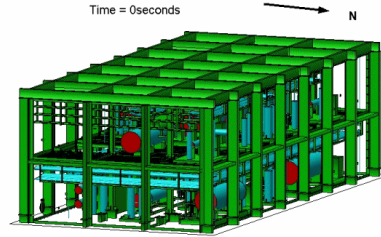
- Simulations (product fraction)



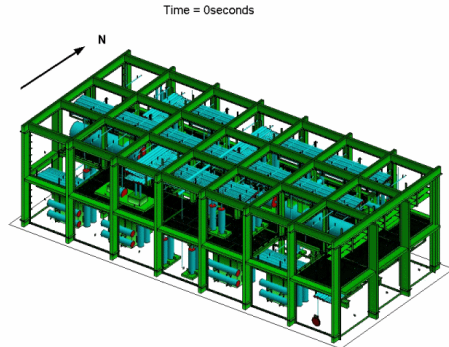
Flame propagation



Spadeadam gas build-up



-up case A30



Spadeadam gas build-up case A26



PARTICIPANTS AT
COURSE NO. I
August 26-27, 1986

1986

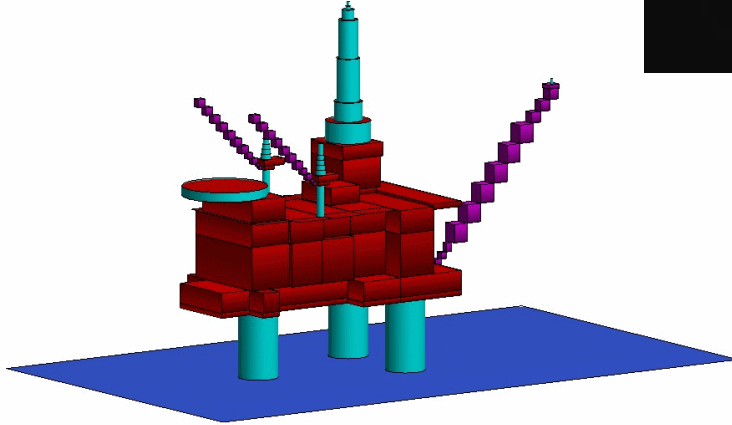
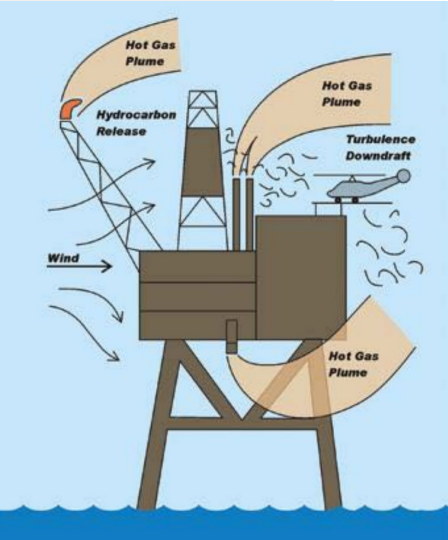
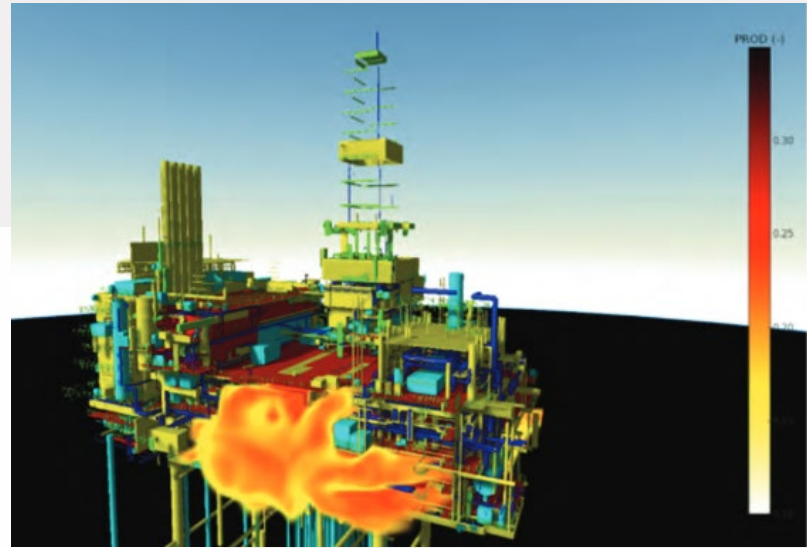
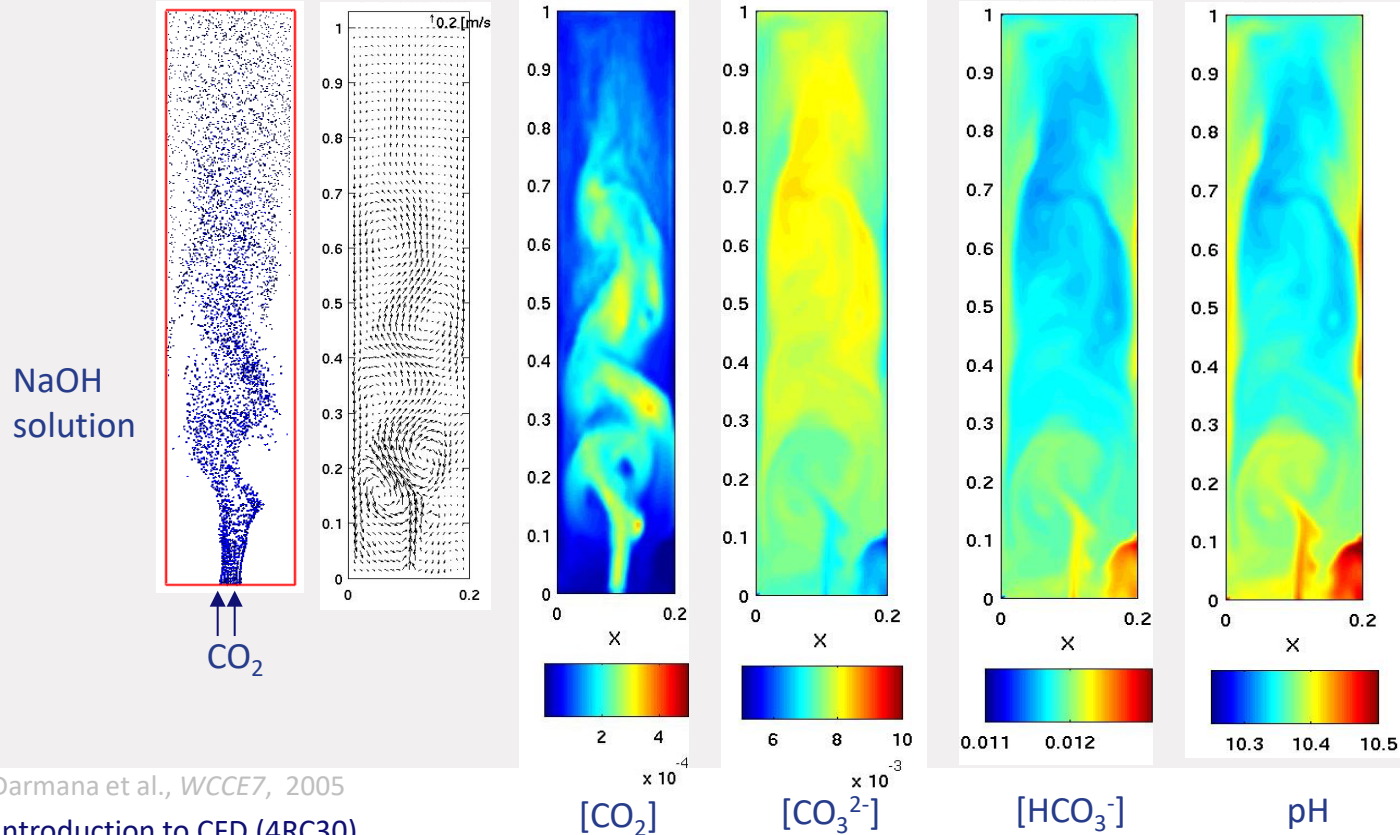


Image credits: Aalborg University Esbjerg
https://nl.wikipedia.org/wiki/Piper_Alpha

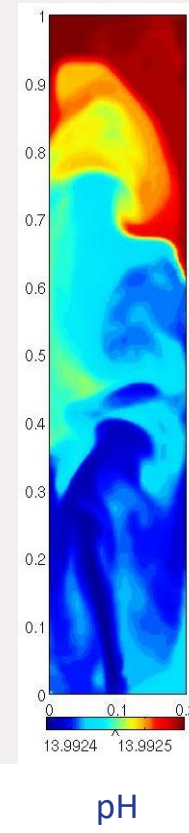
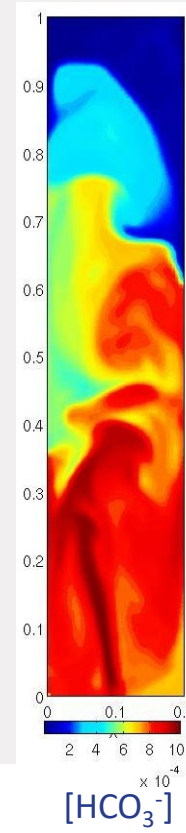
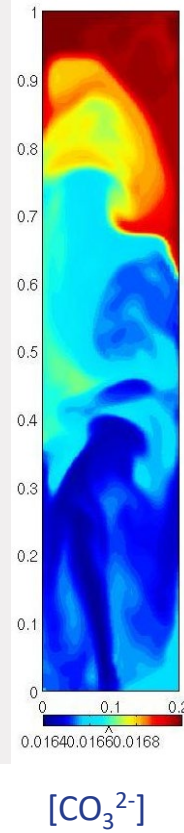
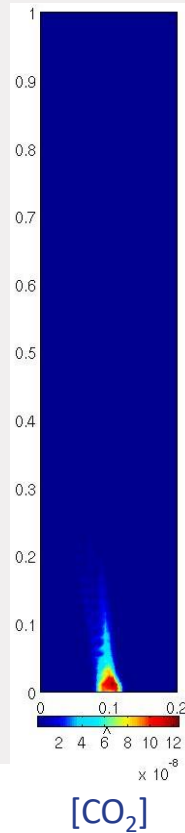
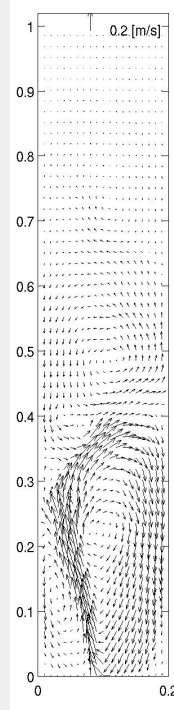
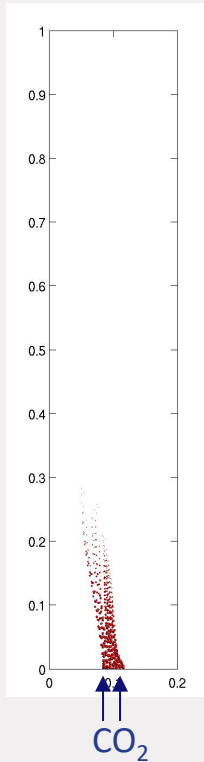
Chemisorption process @ pH = 10.5



Darmana et al., WCCE7, 2005

Chemisorption process @ pH = 14

NaOH
solution



Darmana et al., WCCE7, 2005

Wrap up

Determine slowest timescale

- t_{macro} **simple chemical reacting system:**
PDE for mixture fraction ξ
- t_{micro} **eddy break-up model:**
PDEs for mixture fraction ξ and fuel mass fraction Y_{fu}
speed of mixing in sink S_{fu}
- $t_{micro} \sim t_r$ **eddy break-up model with reaction:**
PDEs for mixture fraction ξ and fuel mass fraction Y_{fu}
speeds of mixing and reaction in sink S_{fu}